

DATA STRUCTURES LAB MANUAL - TEST CASES

LAB PROGRAM 1 — Linear & Binary Search

Test Case ID	Input	Expected Output
TC1	{4,7,3,2,1,7,9}	Element found
TC2	{1,2,3,4,5}	Element not found
TC3	{7,7,7,7}	First occurrence index = 0

LAB PROGRAM 2 — Bubble Sort

Test Case ID	Input	Expected Output
TC1	{5,3,1,6,0,2,4}	{0,1,2,3,4,5,6}
TC2	{9,8,7,6}	{6,7,8,9}
TC3	{5,5,5}	{5,5,5}

LAB PROGRAM 3 — Selection & Insertion Sort

Test Case ID	Input	Expected Output
TC1	{75,8,1,16,48,3,7,0}	{75,48,16,8,7,3,1,0}
TC2	{1,2,3}	{3,2,1}
TC3	{100}	{100}

LAB PROGRAM 4 — Singly Linked List

Test Case ID	Operation	Expected Result
TC1	Insert 61	61 -> NULL
TC2	Delete 61	16 -> NULL
TC3	Delete all nodes	NULL

LAB PROGRAM 5 — Linear Queue

Test Case ID	Operation	Expected Output
TC1	Enqueue 61	Queue contains 61
TC2	Dequeue once	Removes first element
TC3	Queue full	Overflow message

LAB PROGRAM 6 — Circular Queue

Test Case ID	Operation	Expected Result
TC1	Insert 4 elements	Queue full
TC2	Delete all	Queue empty
TC3	Delete empty queue	Underflow

LAB PROGRAM 7 — Ordered Linked List

Test Case ID	Input	Expected Ordered List
TC1	61,16,8,27	8 ->16 ->27 ->61
TC2	Delete 8	16 ->27 ->61
TC3	Delete tail	Tail removed

LAB PROGRAM 8 — Polynomial Addition

Test Case ID	Polynomial 1	Expected Result
TC1	$6x^3+10x^2+5$	$6x^3+14x^2+2x+6$
TC2	$x^2 + x^2$	$2x^2$
TC3	$0 + x+1$	$x+1$

LAB PROGRAM 9 — Stack Operations

Test Case ID	Operation	Expected Output
TC1	Push elements	Elements inserted
TC2	Pop once	Last element removed
TC3	Underflow	Cannot pop

LAB PROGRAM 10 — Recursive GCD

Test Case ID	Input	Expected GCD
TC1	4,6,8	2
TC2	10,20,30	10
TC3	7,11,13	1

LAB PROGRAM 11 — Circular Queue Using Linked List

Test Case ID	Operation	Expected Result
TC1	Insert 5,7,0	Circular structure formed
TC2	Delete 6	Node removed
TC3	Delete rear	Rear updated

LAB PROGRAM 12 — Infix to Postfix

Test Case ID	Infix Expression	Expected Postfix
TC1	$x^y/(5*z)+2$	$xy^5z*/2+$
TC2	$(A+B)*C$	$AB+C*$
TC3	$A+B*C$	$ABC*+$

LAB PROGRAM 13 — Postfix Evaluation

Test Case ID	Postfix	Expected Result
TC1	$53+82-*$	48
TC2	$23+$	5
TC3	$82/$	4

LAB PROGRAM 14 — Binary Tree Operations

Test Case ID	Operation	Expected Result
TC1	Insert nodes	Tree created
TC2	Delete 15	Node removed
TC3	Traversal	Sorted inorder output

LAB PROGRAM 15 — BST Traversals

Test Case ID	Traversal Type	Expected Output
TC1	Inorder	0 1 2 3 5 6 9
TC2	Preorder	2 1 0 5 3 9 6
TC3	Postorder	0 1 3 6 9 5 2

LAB PROGRAM 16 — Heap Sort

Test Case ID	Input	Expected Output
TC1	{9,16,32,8,4,1,5,8,0}	{0,1,4,5,8,8,9,16,32}

TC2	Reverse order	Sorted ascending
TC3	Duplicate elements	Correct sorting

LAB PROGRAM 17 — String Operations

Test Case ID	Operation	Expected Output
TC1	Length of Flowers	7
TC2	Concatenate strings	Flowers are beautiful
TC3	Replace are with is	is beautiful

Additional Viva Questions

1. Difference between Stack and Queue
2. Why Binary Search requires sorted array
3. Time complexity of Bubble Sort
4. Difference between Linear Queue and Circular Queue
5. Applications of Linked List
6. What is recursion?
7. Difference between BST and Binary Tree
8. Heap Sort complexity
9. Prefix vs Postfix expression
10. Advantages of Dynamic Memory Allocation